

Saddle-Field Policy Optimization for Zero-Sum Markov Games: Adaptive Lyapunov-Drift Control with Finite-Time Stationarity Guarantees

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Abstract—Two-player zero-sum Markov games model robust reinforcement learning and competitive policy optimization. Rotational components in their saddle fields can hinder first-order updates. We identify the policy field $F(z)$ and a Jacobian-induced curvature direction $G(z) = \nabla F(z)F(z)$ for rotational coupling. A symmetric–antisymmetric Jacobian decomposition characterizes when the curvature direction decreases field energy. We then develop a geometry-aware controller that assigns separate step sizes to the two directions. Minimizing a local model of the one-step change in a composite Lyapunov function $\mathcal{V}$ yields the two step sizes through a box-constrained quadratic program (QP). Under local geometric coverage, the controller attains an $O(1/K)$ bound over K joint updates on the time-average expected stationarity residual, plus an explicit residual floor. A positive-semidefinite symmetric part and nonsingularity of the metric-transformed Jacobian at the target stationary point are shown to imply neighborhood coverage, including the nonsingular pure-rotation case. Under a local policy-space error bound, this stationarity result transfers to guarantees on the unregularized Nash gap and the maximizing player’s worst-case-return loss. Experiments across four settings validate the predicted geometry; at the final checkpoint of every Markov-game task, the controller has higher mean worst-case return and lower mean exploitability than the matched field-only ablation, including a 79.1% exploitability reduction on neural CyclicalControl.

Index Terms—Zero-sum Markov games, robust reinforcement learning, stochastic saddle fields, Lyapunov drift, adaptive step size.

I. INTRODUCTION

TWO-PLAYER zero-sum Markov games provide a mathematical model for competitive sequential decisions [1]–[3]. They also underpin robust reinforcement learning (RL), where a maximizing policy must retain performance against model uncertainty, disturbances, or adversarial actions represented by a minimizing policy [4]–[9].

Policy parameterization yields joint ascent–descent dynamics estimated from stochastic trajectories. Unlike an ordinary minimization gradient, the resulting saddle field may contain a strong rotational component that makes gradient descent–ascent (GDA) cycle or diverge even on bilinear games; related instability arises in robust-control formulations of adversarial RL [10].

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The proximal point method (PPM) and extragradient-type methods, including the classical extragradient method (EGM), Mirror-Prox, and policy extragradient, introduce corrections that can suppress rotational dynamics at suitable step sizes [11]–[15]. Recent analyses of saddle algorithms relate these corrections to Jacobian-induced dynamics in the underlying game field [16]–[20]. Adaptive variational-inequality and extra-gradient methods adjust the extrapolation and update scales from observed operator information [21], [22]. Our controller instead treats the field and Jacobian-induced curvature directions as separate search coordinates and selects their step sizes by minimizing a local model of composite Lyapunov change.

We represent the parameterized game as a saddle field with two complementary update directions. The saddle field $F(z)$ defines the joint first-order update direction $-F(z)$, while the curvature direction $+G(z) = \nabla F(z)F(z)$ captures the local effect of rotational coupling. An exact symmetric–antisymmetric Jacobian condition shows that $+G$ decreases the squared field norm precisely when the antisymmetric action dominates the symmetric action along F .

Because the effective balance also changes along the trajectory and under stochastic noise, we extend Lyapunov-drift step-size design to select both magnitudes online using their local geometry [23], [24].

The resulting controller minimizes a local second-order model of a composite Lyapunov function through a two-dimensional box-constrained quadratic program (QP), yielding closed-form step sizes, finite-time stationarity guarantees, and policy-space performance bounds.

The main contributions are as follows.

- **Curvature geometry.** We derive an exact symmetric–antisymmetric Jacobian criterion for curvature-induced field-energy descent and connect it to the correction shared by PPM and EGM but absent from GDA.
- **Adaptive Lyapunov-drift control.** A composite merit combining weighted field energy and a regularized policy-space gap yields separate field and curvature step sizes through a closed-form box QP. Its local model dominates every feasible field-only, curvature-only, or fixed-coupling comparator in the same box, with strict interior improvement characterized by a reduced gradient.
- **Finite-time and policy-space guarantees.** For biased stochastic oracles, two-direction geometric coverage yields an $O(1/K)$ transient in the time-average expected stationarity residual plus an explicit error floor. We derive neighborhood coverage from a positive-semidefinite symmetric part and

nonsingularity of the metric-transformed Jacobian at the target, including pure rotation; under a local policy-space error bound, the stationarity estimate transfers to ergodic Nash-gap and worst-case-return bounds.

II. SADDLE-FIELD DYNAMICS IN ZERO-SUM MARKOV GAMES

A. Zero-Sum Markov Game and Saddle-Field Formulation

We consider an infinite-horizon discounted two-player zero-sum Markov game with state space $\mathcal{S}$, maximizing-player action space $\mathcal{A}$, minimizing-player action space $\mathcal{B}$, transition kernel P , bounded reward r to the maximizing player, discount factor $\alpha \in (0, 1)$, and initial-state distribution ρ . For a maximizing-player policy π and a minimizing-player policy ν , the maximizing player's discounted return is

$$J(\pi, \nu) \triangleq \mathbb{E}_{\rho, \pi, \nu} [\sum_{t=0}^{\infty} \alpha^t r(s_t, a_t, b_t)].$$

Here $\mathbb{E}_{\rho, \pi, \nu}$ denotes expectation under the initial-state distribution and the two policies. The policy-space saddle objective is $\sup_{\pi} \inf_{\nu} J(\pi, \nu)$. When $\mathcal{A}$ and $\mathcal{B}$ are finite, define the discounted causal entropies

$$\begin{aligned} H_{\alpha}^{\pi}(\pi, \nu) &\triangleq \mathbb{E}_{\rho, \pi, \nu} \left[\sum_{t \geq 0} \alpha^t (-\log \pi(a_t | s_t)) \right], \\ H_{\alpha}^{\nu}(\pi, \nu) &\triangleq \mathbb{E}_{\rho, \pi, \nu} \left[\sum_{t \geq 0} \alpha^t (-\log \nu(b_t | s_t)) \right], \end{aligned}$$

with $0 \log 0 = 0$. For entropy weight $\varepsilon \geq 0$, the corresponding regularized return is

$$J_{\varepsilon}(\pi, \nu) \triangleq J(\pi, \nu) + \varepsilon H_{\alpha}^{\pi}(\pi, \nu) - \varepsilon H_{\alpha}^{\nu}(\pi, \nu). \quad (1)$$

The opposite entropy signs follow the saddle convention: the maximizing player raises J_{ε} by increasing its entropy, while the minimizing player lowers J_{ε} by increasing its own entropy. Thus, entropy encourages stochastic responses for both players and replaces potentially set-valued hard best responses by soft best responses in finite games. Without regularization, the supremum or infimum defining a best-response value can kink when its optimizer switches, invalidating the gradient-Hessian quadratic model used by the controller. For $\varepsilon > 0$, this smoothing supplies the differentiable policy-space gap needed by the first- and second-order Lyapunov model; Proposition 8 formalizes the required local regularity. Because regularization perturbs the original performance criterion, Proposition 7 later transfers the regularized guarantee to the unregularized Nash gap and worst-case return with an explicit entropy-bias term. Let $J_{\text{alg}} \in \{J, J_{\varepsilon}\}$ denote the differentiable saddle objective. This notation separates the objective differentiated by the controller from the performance measure used for final evaluation. The finite-game controller and all Markov-game experiments use $J_{\text{alg}} = J_{\varepsilon}$, while Section IV-C reserves J for unregularized performance measures. In large finite or continuous state spaces, we adopt the standard parameterized formulation because exact policy-space optimization is generally intractable. The maximizing-player policy is represented by $\pi_{\theta}(a | s)$ with

$\theta \in \mathbb{R}^{d_{\theta}}$, and the minimizing-player policy by $\nu_{\psi}(b | s)$ with $\psi \in \mathbb{R}^{d_{\psi}}$, giving

$$J_{\text{alg}}(\theta, \psi) \triangleq J_{\text{alg}}(\pi_{\theta}, \nu_{\psi}),$$

$$\max_{\theta} \min_{\psi} J_{\text{alg}}(\theta, \psi).$$

Write $z \triangleq (\theta, \psi) \in \mathbb{R}^d$, where $d \triangleq d_{\theta} + d_{\psi}$. The joint policy ascent-descent dynamics define the saddle field

$$F(z) \triangleq \begin{bmatrix} -\nabla_{\theta} J_{\text{alg}}(\theta, \psi) \\ \nabla_{\psi} J_{\text{alg}}(\theta, \psi) \end{bmatrix}. \quad (2)$$

The first-order stationary set is $\mathcal{Z}^* \triangleq \{z : F(z) = \mathbf{0}\}$, where $\mathbf{0}$ is the zero vector. Every $z^* \in \mathcal{Z}^*$ satisfies the unconstrained first-order Karush-Kuhn-Tucker (KKT) conditions for the parameterized zero-sum objective. Under neural policy parameterization, $\mathcal{Z}^*$ is the parameter-space first-order stationary set. This distinction matters because a nonconvex or noninjective policy parameterization can satisfy $F(z) = \mathbf{0}$ even when a player can still improve by taking a best response in the full policy space. The policy-space results in Section IV-C use the Nash gap to measure this remaining unilateral-improvement incentive and additionally bound the maximizing player's worst-case-return loss.

B. Curvature-Induced Dynamics and Jacobian Geometry

The saddle-field formulation exposes the rotational component that can destabilize first-order updates. To understand how established saddle methods counter this effect, we examine the Jacobian-based correction generated by PPM and EGM. For a scalar step size $s > 0$, GDA, PPM, and EGM use the representative updates

$$\begin{aligned} \text{GDA: } z_+^{\text{GDA}} &= z - sF(z), \\ \text{PPM: } z_+^{\text{PPM}} &= z - sF(z_+^{\text{PPM}}), \\ \text{EGM: } \tilde{z} &= z - sF(z), \quad z_+^{\text{EGM}} = z - sF(\tilde{z}), \end{aligned} \quad (3)$$

where $\tilde{z}$ is the EGM extrapolation point. GDA is explicit, PPM is implicit, and EGM uses an extrapolated field query. Below, $\|\cdot\|$ denotes the Euclidean norm and its induced operator and tensor norms, $\langle \cdot, \cdot \rangle$ the standard inner product, ∇F the Jacobian of F with respect to z , and $\nabla^2 F$ its second-derivative tensor. We write I_n for the $n \times n$ identity matrix and $\mathbf{I}$ when its dimension is clear; for a square matrix X , $\text{sym}(X) \triangleq (X + X^{\top})/2$ and $\text{skew}(X) \triangleq (X - X^{\top})/2$, while $X \succeq 0$ and $X \succ 0$ respectively denote positive semidefiniteness and positive definiteness when X is symmetric.

The following assumption states the local regularity required by the template expansion and one-step analysis.

Assumption 1 (Local smoothness). *There exists an open set $\mathcal{D} \subset \mathbb{R}^d$ such that: (a) all iterates and probing points used in the analysis lie in $\mathcal{D}$; (b) $J_{\text{alg}} : \mathbb{R}^d \rightarrow \mathbb{R}$ is four times continuously differentiable (C^4) on $\mathcal{D}$; (c) all derivatives of J_{alg} up to order four are bounded on $\mathcal{D}$.*

Conditions (b)–(c) imply that F is three times continuously differentiable, i.e. $F \in C^3(\mathcal{D}; \mathbb{R}^d)$; after prescribing an implementable region $\mathcal{T} \subset \mathcal{D}$, the safeguards of Section V can maintain Condition (a).

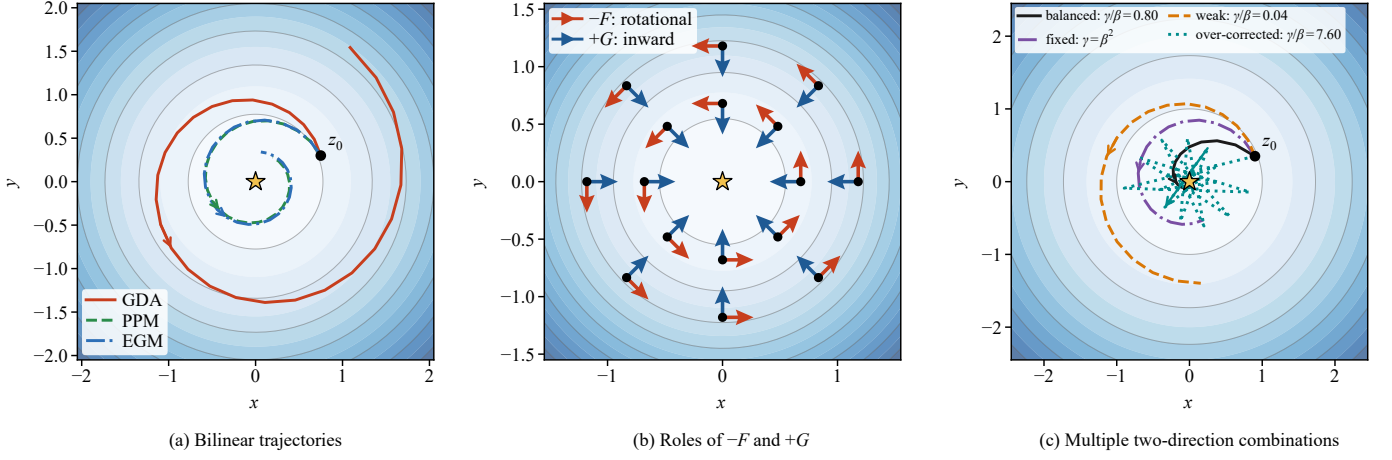


Fig. 1. Geometry of two-direction saddle updates. (a) GDA spirals outward, whereas PPM and EGM contract at the displayed step size. (b) On the bilinear game, the first-order field step is tangential to the residual contours, whereas the shared second-order correction points inward. (c) The curvature-to-field step-size ratio controls contraction and residual rotation.

Proposition 1 (Discrete-time second-order template expansions). *Suppose Assumption 1 holds and fix $z \in \mathcal{D}$. There exists $\bar{s} > 0$ such that, for every $s \in (0, \bar{s}]$, the PPM equation has a unique local solution, the EGM extrapolation and update points lie in $\mathcal{D}$, and*

$$\text{GDA: } z_+ - z = -sF(z),$$

$$\text{PPM: } z_+ - z = -sF(z) + s^2 \nabla F(z) F(z) + R_s^{\text{PPM}}(z),$$

$$\text{EGM: } z_+ - z = -sF(z) + s^2 \nabla F(z) F(z) + R_s^{\text{EGM}}(z),$$

with $\|R_s^{\text{PPM}}(z)\| \leq C_{\text{PPM}} s^3$ and $\|R_s^{\text{EGM}}(z)\| \leq C_{\text{EGM}} s^3$ for finite constants depending only on local bounds of F , ∇F , and $\nabla^2 F$ on $\mathcal{D}$.

Unlike GDA, PPM and EGM share the leading-order correction $+s^2 \nabla F(z) F(z)$, which can suppress rotational saddle dynamics at suitable step sizes. This contrast motivates writing $A(z) \triangleq \nabla F(z)$ and isolating the curvature-induced direction

$$G(z) \triangleq A(z)F(z). \quad (4)$$

For any fixed symmetric positive-definite metric M , define the metric field energy $\mathcal{E}_M(z) \triangleq \frac{1}{2} F(z)^\top M F(z)$ and its exact-expectation (population) decrease signals

$$\chi_F^M(z) \triangleq -\langle \nabla \mathcal{E}_M(z), -F(z) \rangle = F^\top \text{sym}(MA)F,$$

$$\chi_G^M(z) \triangleq -\langle \nabla \mathcal{E}_M(z), G(z) \rangle = -F^\top MA^2 F. \quad (5)$$

Positive values certify first-order field-energy decrease along the corresponding directions. The Euclidean choice $M = I$ gives $\phi(z) \triangleq \mathcal{E}_I(z) = \frac{1}{2} \|F(z)\|^2$, which is nonnegative and vanishes exactly on $\mathcal{Z}^*$.

We first illustrate this geometric criterion on the bilinear max-min saddle $L(x, y) = -xy$, which has field $F = J_0 z$ with

$$J_0 \triangleq \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

We denote the nonnegative step sizes along $-F$ and $+G$ by β and γ , respectively. Fig. 1 uses this bilinear saddle to connect the behavior of the classical updates with the separate control of the field and curvature directions. Panel (a) compares the update trajectories: GDA has eigenvalue modulus $\sqrt{1+s^2} > 1$ and spirals outward, PPM contracts for every

$s > 0$, and EGM contracts for $0 < s < 1$ (is neutral at $s = 1$ and expansive for $s > 1$). Panel (b) explains this contrast geometrically. Since $J_0^2 = -I_2$, $G = J_0^2 z = -z = -\nabla \phi$ points inward, whereas $-F$ is tangent to the field-energy level sets; hence, the shared $+s^2 G$ correction enables PPM and EGM to counter the rotational behavior that destabilizes GDA in this example. Panel (c) compares four field-curvature step-size combinations. Proposition 1 and Eq. (4) show that PPM and EGM couple these step sizes as (s, s^2) to second order, whereas the distinct trajectories in panel (c) motivate selecting β and γ independently.

To characterize when $+G$ decreases the Euclidean field energy, fix $z \in \mathcal{D}$, write $A = A(z)$, and split it into its symmetric and antisymmetric parts

$$A = S + W, \quad S = \text{sym}(A), \quad W = \text{skew}(A).$$

The symmetric part S captures nonrotational dynamics, whereas the antisymmetric part W captures rotational dynamics [17]. The following lemma expresses the field-energy directional derivatives exactly in terms of S and W .

Lemma 1 (Symmetric-antisymmetric decomposition of the field-energy derivatives). *Let $\phi(z) = \frac{1}{2} \|F(z)\|^2$ and $A = \nabla F(z)$, so that $\nabla \phi(z) = A^\top F(z)$. The directional derivatives of ϕ along $-F$ and $+G = AF$ satisfy*

$$\langle \nabla \phi(z), -F(z) \rangle = -F(z)^\top S F(z), \quad (6)$$

$$\langle \nabla \phi(z), G(z) \rangle = \|S F(z)\|^2 - \|W F(z)\|^2. \quad (7)$$

The compact form in Eq. (7) uses the cancellation $F^\top (SW + WS) F = 0$, since $SW + WS$ is antisymmetric. Eq. (6) shows that the field direction decreases field energy through the symmetric part: in a purely rotational field ($S = 0$), $-F$ is tangent to the level sets of ϕ , exactly the GDA cycling mechanism. Eq. (7) balances the symmetric action $\|S F\|^2$ against the antisymmetric action $\|W F\|^2$ and yields the following skew-dominance criterion.

Theorem 1 (Skew-dominance characterization of curvature descent). *Fix $z \in \mathcal{D}$ with $F(z) \neq 0$, and define $A = \nabla F(z)$, $S = \text{sym}(A)$, and $W = \text{skew}(A)$. The curvature direction $+G(z)$ is a strict descent direction for the field energy ϕ at*

$\mathbf{z}$, i.e. $\langle \nabla \phi(\mathbf{z}), \mathbf{G}(\mathbf{z}) \rangle < 0$, if and only if the antisymmetric action dominates the symmetric action along $\mathbf{F}$:

$$\|\mathbf{WF}(\mathbf{z})\| > \|\mathbf{SF}(\mathbf{z})\|. \quad (8)$$

In particular:

- (i) if the field is purely rotational at $\mathbf{z}$ ($\mathbf{SF}(\mathbf{z}) = \mathbf{0}$ and $\mathbf{WF}(\mathbf{z}) \neq \mathbf{0}$), then $\langle \nabla \phi, -\mathbf{F} \rangle = 0$ and $\langle \nabla \phi, \mathbf{G} \rangle = -\|\mathbf{WF}(\mathbf{z})\|^2 < 0$; hence $+\mathbf{G}$ strictly decreases ϕ , whereas $-\mathbf{F}$ is tangential and has zero first-order effect;
- (ii) if the field is symmetric at $\mathbf{z}$ ($\mathbf{W} = \mathbf{0}$), then $\langle \nabla \phi, \mathbf{G} \rangle = \|\mathbf{SF}(\mathbf{z})\|^2 \geq 0$; hence $+\mathbf{G}$ preserves or increases ϕ to first order;
- (iii) if $\mathbf{A}$ is normal ($\mathbf{A}^\top \mathbf{A} = \mathbf{A} \mathbf{A}^\top$) and its restriction to a two-dimensional real invariant subspace $\mathcal{U}$ has the orthonormal representation $\mathbf{A}|_{\mathcal{U}} = \mu \mathbf{I}_2 + \sigma J_0$, with conjugate eigenvalues $\mu \pm i\sigma$ ($i^2 = -1$) and $\mathbf{F}(\mathbf{z}) \in \mathcal{U}$, then the condition in Eq. (8) reduces to $|\sigma| > |\mu|$.

This theorem provides the geometric criterion that motivates the two-direction controller developed in Section III. In part (iii), μ and σ are the real and imaginary parts of the conjugate eigenvalues: for a local coordinate $\mathbf{u} \in \mathcal{U}$, the linearized flow $\dot{\mathbf{u}} = -(\mu \mathbf{I}_2 + \sigma J_0)\mathbf{u}$ shows that they control radial scaling and oriented angular motion, respectively. Moreover, $\mathbf{S}|_{\mathcal{U}} = \mu \mathbf{I}_2$ and $\mathbf{W}|_{\mathcal{U}} = \sigma J_0$, so $\|\mathbf{SF}\| = |\mu| \|\mathbf{F}\|$ and $\|\mathbf{WF}\| = |\sigma| \|\mathbf{F}\|$ on $\mathcal{U}$. Part (iii) is an interpretable normal-block specialization, not a structural assumption imposed on the general saddle field or on the controller. Jacobian–vector and Jacobian–transpose–vector products evaluate this condition and yield the measures $\|\mathbf{WF}\|$ and $\|\mathbf{SF}\|$ used in Section VI.

For the local linear field $\mathbf{F}(\mathbf{z}) = \mathbf{A}(\mathbf{z} - \mathbf{z}^*)$ at $\mathbf{F}(\mathbf{z}) \neq \mathbf{0}$, $\mathbf{G} = \mathbf{AF}$ is collinear with $\mathbf{F}$ exactly when $\mathbf{F}$ is a real eigenvector of $\mathbf{A}$. If also $\mathbf{A}^\top = -\mathbf{A}$, $\mathbf{A}^2 = -\sigma^2 \mathbf{I}$, and $\sigma \neq 0$, then for $\mathbf{u} = \mathbf{z} - \mathbf{z}^* \neq \mathbf{0}$, $\mathbf{F} = \mathbf{Au}$ is tangential and $\mathbf{G} = -\sigma^2 \mathbf{u}$ is orthogonally inward, with $\langle \nabla \phi, -\mathbf{F} \rangle = 0$ and $\langle \nabla \phi, \mathbf{G} \rangle = -\sigma^4 \|\mathbf{u}\|^2 < 0$.

C. From Classical Templates to a Stochastic Recursion

Our aim is a simultaneous stochastic policy update for zero-sum Markov games that adapts to rotational coupling. Relative to GDA's step $-s\mathbf{F}$, Proposition 1 shows that PPM and EGM add $+s^2\mathbf{G} + O(s^3)$; in the rotation-dominated regimes characterized above, this curvature action suppresses the rotational behavior of the field step. Accordingly, at joint update k , let $\hat{\mathbf{F}}_k$ and $\hat{\mathbf{G}}_k$ denote the observed field and curvature directions and let $\beta_k, \gamma_k \geq 0$ denote their step sizes; the stochastic two-direction recursion is

$$\mathbf{z}_{k+1} = \mathbf{z}_k - \beta_k \hat{\mathbf{F}}_k + \gamma_k \hat{\mathbf{G}}_k, \quad (9)$$

With exact population directions, $(s, 0)$ gives GDA and (s, s^2) matches PPM/EGM to second order; finite-trajectory policy optimization instead uses stochastic observations [25]. For the stochastic analysis, let $\{\mathcal{F}_k\}_{k \geq 0}$ be the filtration generated by the algorithmic history revealed before the queries at joint update k .

Assumption 2 (Biased stochastic oracles). *For every k , let $\mathbf{b}_k^F, \mathbf{b}_k^G$ be $\mathcal{F}_k$ -measurable oracle biases and let $\boldsymbol{\xi}_k^F, \boldsymbol{\xi}_k^G$ be*

conditionally zero-mean noise variables, i.e. $\mathbb{E}[\boldsymbol{\xi}_k^F | \mathcal{F}_k] = \mathbb{E}[\boldsymbol{\xi}_k^G | \mathcal{F}_k] = \mathbf{0}$. The observed directions decompose as

$$\hat{\mathbf{F}}_k = \mathbf{F}(\mathbf{z}_k) + \mathbf{b}_k^F + \boldsymbol{\xi}_k^F, \quad \hat{\mathbf{G}}_k = \mathbf{G}(\mathbf{z}_k) + \mathbf{b}_k^G + \boldsymbol{\xi}_k^G,$$

There exist finite $\mathcal{F}_k$ -measurable scalars $\bar{b}_k^F, \bar{b}_k^G, \sigma_{2,k}, \sigma_{3,k} \geq 0$ such that, almost surely,

- (a) $\|\mathbf{b}_k^F\| \leq \bar{b}_k^F$ and $\|\mathbf{b}_k^G\| \leq \bar{b}_k^G$;
- (b) $\mathbb{E}[\|\boldsymbol{\xi}_k^F\|^2 + \|\boldsymbol{\xi}_k^G\|^2 | \mathcal{F}_k] \leq \sigma_{2,k}^2$;
- (c) $\mathbb{E}[\|\hat{\mathbf{F}}_k\|^3 + \|\hat{\mathbf{G}}_k\|^3 | \mathcal{F}_k] \leq \sigma_{3,k}^3$.

Here $\bar{b}_k^F, \bar{b}_k^G$ are bias bounds, while $\sigma_{2,k}, \sigma_{3,k}$ bound the conditional second noise moment and third oracle moment. This model covers unbiased mini-batch estimators and bounded bias from truncated trajectories, clipped surrogates, or approximate curvature actions.

III. GEOMETRY-AWARE ADAPTIVE LYAPUNOV-DRIFT CONTROL

Section II shows that the field and curvature directions have different local effects on field energy. We now encode stationarity and policy performance in a nonnegative Lyapunov function. This extension is necessary because field energy alone cannot reveal whether a locally stabilizing parameter update improves policy-space performance. We then distinguish its realized one-step change from its conditional Lyapunov drift and construct an implementable local model for selecting the two step sizes.

A. Lyapunov Merit Function

The field energy vanishes with the parameter-space stationarity residual, but it does not by itself rule out a parameterization-induced stationary point with positive policy-space exploitability. Conversely, the Nash gap measures unilateral policy improvement but neither directly measures the local saddle-field residual nor naturally shares its numerical scale. The following merit function therefore combines the two quantities: field energy guides local optimization, while the regularized gap retains policy-space performance in the step-size decision.

Definition 1 (Composite Lyapunov function). *Let $\text{diag}(\cdot)$ denote block-diagonal concatenation and let $\eta_\psi > 0$ be the relative weight of the minimizing-player gradient block. Define the block-diagonal weight matrix*

$$\mathbf{W}_V \triangleq \text{diag}(\mathbf{I}_{d_\theta}, \eta_\psi \mathbf{I}_{d_\psi}) \in \mathbb{R}^{d \times d},$$

The weighted field-energy component is $\mathcal{E}_F(\mathbf{z}) \triangleq \mathcal{E}_{\mathbf{W}_V}(\mathbf{z}) = \frac{1}{2} \mathbf{F}(\mathbf{z})^\top \mathbf{W}_V \mathbf{F}(\mathbf{z})$. Henceforth, $\chi_F \triangleq \chi_F^{\mathbf{W}_V}$ and $\chi_G \triangleq \chi_G^{\mathbf{W}_V}$ denote the corresponding population decrease signals already defined in Eq. (5). For finite games, fix $\varepsilon > 0$, write $J_\varepsilon = J_\varepsilon(\pi_\theta, \nu_\psi)$, and define the performance component

$$\begin{aligned} \mathcal{P}(\mathbf{z}) &\triangleq \text{Gap}_\varepsilon(\pi_\theta, \nu_\psi) \\ &= \left[\sup_{\bar{\pi}} J_\varepsilon(\bar{\pi}, \nu_\psi) - J_\varepsilon \right] + \left[J_\varepsilon - \inf_{\bar{\nu}} J_\varepsilon(\pi_\theta, \bar{\nu}) \right], \end{aligned}$$

The extrema range over stationary randomized policies; Eq. (1) defines J_ε . The first bracket is the return improvement available to the maximizing player through a regularized best response,

and the second is the return reduction available to the minimizing player, which seeks a smaller J_ε . Both gains are nonnegative, so $\mathcal{P}(z) \geq 0$, and $\mathcal{P}(z) = 0$ exactly at a regularized policy-space Nash equilibrium, where neither player can improve unilaterally. When this finite-game component is unavailable, taking $\mathcal{P} \equiv 0$ and $\lambda_P = 0$ recovers field-energy-only control. For fixed normalizers $\kappa_F, \kappa_P > 0$ and weights $\lambda_F > 0$, $\lambda_P \geq 0$, define the composite Lyapunov function

$$\mathcal{V}(z) \triangleq \lambda_F \frac{\mathcal{E}_F(z)}{\kappa_F} + \lambda_P \frac{\mathcal{P}(z)}{\kappa_P}. \quad (10)$$

The normalizers place components with different numerical scales on a comparable basis, while the λ weights specify their relative influence on the controller.

For the recursion in Eq. (9), define the realized one-step Lyapunov increment by $\mathcal{J}_k \triangleq \mathcal{V}(z_{k+1}) - \mathcal{V}(z_k)$. Its conditional expectation

$$\mathcal{D}_k \triangleq \mathbb{E}[\mathcal{J}_k \mid \mathcal{F}_k]$$

is the Lyapunov drift at update k . The increment $\mathcal{J}_k$ is pathwise after the current stochastic queries are realized, whereas $\mathcal{D}_k$ averages this increment conditional on the pre-query history $\mathcal{F}_k$. Equivalently, $\mathcal{D}_k < 0$ means that the next update decreases the composite merit $\mathcal{V}$ in conditional expectation; a more negative drift represents a larger expected one-step reduction in the combined stationarity and policy-performance measure. This interpretation makes minimizing $\mathcal{D}_k$ the ideal step-size design objective.

The block weight $\mathbf{W}_V$ is unrelated to the antisymmetric Jacobian part $\mathbf{W} = \text{skew}(\nabla \mathbf{F})$.

Remark 1 (From field-energy geometry to Lyapunov control). Theorem 1 analyzes $+\mathbf{G}$ for the field energy ϕ ; Definition 1 extends the same geometry to $\mathcal{V}$, whose local drift model determines how strongly curvature is combined with policy performance.

Lemma 2 (Properties of the composite Lyapunov function). Suppose Assumption 1 holds and $\mathcal{P}$ is nonnegative and C^3 on $\mathcal{D}$. Then: (i) $\mathcal{V}(z) \geq 0$ on $\mathcal{D}$, and since $\lambda_F > 0$, $\mathcal{V}(z) = 0$ implies $z \in \mathcal{Z}^*$; (ii) $\mathcal{V} \in C^3(\mathcal{D})$; (iii) the gradient is $\nabla \mathcal{V} = (\lambda_F/\kappa_F)\nabla \mathcal{E}_F + (\lambda_P/\kappa_P)\nabla \mathcal{P}$, where $\nabla \mathcal{E}_F = \nabla \mathbf{F}^\top \mathbf{W}_V \mathbf{F}$; (iv) $\Phi(z) \triangleq \mathbf{F}(z)^\top \mathbf{W}_V \mathbf{F}(z)$ is the weighted stationarity residual and satisfies $\Phi(z) \leq 2\kappa_F \mathcal{V}(z)/\lambda_F$.

The composite Lyapunov decrease signals along the two population directions are

$$\begin{aligned} D_F(z) &\triangleq \langle \nabla \mathcal{V}(z), \mathbf{F}(z) \rangle \\ &= \frac{\lambda_F}{\kappa_F} \chi_F(z) + \frac{\lambda_P}{\kappa_P} \langle \nabla \mathcal{P}(z), \mathbf{F}(z) \rangle, \\ D_G(z) &\triangleq -\langle \nabla \mathcal{V}(z), \mathbf{G}(z) \rangle \\ &= \frac{\lambda_F}{\kappa_F} \chi_G(z) - \frac{\lambda_P}{\kappa_P} \langle \nabla \mathcal{P}(z), \mathbf{G}(z) \rangle. \end{aligned} \quad (11)$$

Thus, $D_F > 0$ and $D_G > 0$ mean first-order decrease of $\mathcal{V}$ along $-\mathbf{F}$ and $+\mathbf{G}$, respectively.

B. Second-Order Surrogate for the One-Step Change

For a realized oracle pair and candidate step sizes (β, γ) , define the trial displacement $\Delta_k(\beta, \gamma) \triangleq -\beta \hat{\mathbf{F}}_k + \gamma \hat{\mathbf{G}}_k$ and

the pathwise trial increment $\mathcal{J}_k(\beta, \gamma) \triangleq \mathcal{V}(z_k + \Delta_k(\beta, \gamma)) - \mathcal{V}(z_k)$. The actual quantities satisfy $\mathcal{J}_k = \mathcal{J}_k(\beta_k, \gamma_k)$ and $\mathcal{D}_k = \mathbb{E}[\mathcal{J}_k(\beta_k, \gamma_k) \mid \mathcal{F}_k]$. Constructing a query-dependent rule that exactly minimizes $\mathcal{D}_k$ would require the full conditional oracle distribution and repeated evaluation of the nonlinear post-update function $\mathcal{V}(z_k + \Delta_k)$, neither of which is available from a single query. We therefore minimize, after observing $(\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k)$, a second-order pathwise surrogate of $\mathcal{J}_k(\beta, \gamma)$. The surrogate exposes the separate field curvature, curvature-direction curvature, and cross-curvature needed to select (β_k, γ_k) , while Section IV bounds its mismatch from the realized increment and converts the pathwise choice into an expected finite-time guarantee.

Assumption 3 (Local regularity along the iterates). For every k and every feasible (β, γ) produced by the step-size routine, the segment $\{z_k + t\Delta_k(\beta, \gamma) : t \in [0, 1]\} \subset \mathcal{D}$. There exist $\mathcal{F}_k$ -measurable $G_{V,k}, H_{V,k}, M_{3,k} \geq 0$ that, simultaneously for every feasible pair and almost every realization of the current stochastic queries, satisfy $\|\nabla \mathcal{V}(z_k)\| \leq G_{V,k}$, $\|\nabla^2 \mathcal{V}(z_k)\| \leq H_{V,k}$, and $\sup_{t \in [0, 1]} \|\nabla^3 \mathcal{V}(z_k + t\Delta_k(\beta, \gamma))\| \leq M_{3,k}$.

Lemma 3 (Second-order expansion of the one-step change in $\mathcal{V}$). Under Assumption 3, for every feasible (β, γ) ,

$$\begin{aligned} &\mathcal{V}(z_k + \Delta_k(\beta, \gamma)) - \mathcal{V}(z_k) \\ &= -\beta \langle \nabla \mathcal{V}(z_k), \hat{\mathbf{F}}_k \rangle + \gamma \langle \nabla \mathcal{V}(z_k), \hat{\mathbf{G}}_k \rangle \\ &\quad + \frac{1}{2} \beta^2 \hat{\mathbf{F}}_k^\top \nabla^2 \mathcal{V}(z_k) \hat{\mathbf{F}}_k - \beta \gamma \hat{\mathbf{F}}_k^\top \nabla^2 \mathcal{V}(z_k) \hat{\mathbf{G}}_k \\ &\quad + \frac{1}{2} \gamma^2 \hat{\mathbf{G}}_k^\top \nabla^2 \mathcal{V}(z_k) \hat{\mathbf{G}}_k + \text{rem}_k(\beta, \gamma), \end{aligned} \quad (12)$$

with $|\text{rem}_k(\beta, \gamma)| \leq \frac{M_{3,k}}{6} (\beta \|\hat{\mathbf{F}}_k\| + \gamma \|\hat{\mathbf{G}}_k\|)^3$.

Eq. (12) is the pathwise second-order Taylor expansion for each realized oracle pair. Its linear and quadratic terms define the five scalar coefficients

$$\begin{aligned} e_k &\triangleq \langle \nabla \mathcal{V}(z_k), \hat{\mathbf{F}}_k \rangle, & d_k &\triangleq -\langle \nabla \mathcal{V}(z_k), \hat{\mathbf{G}}_k \rangle, \\ c_k &\triangleq \hat{\mathbf{F}}_k^\top \nabla^2 \mathcal{V}(z_k) \hat{\mathbf{F}}_k, & a_k &\triangleq \hat{\mathbf{G}}_k^\top \nabla^2 \mathcal{V}(z_k) \hat{\mathbf{G}}_k, \\ b_k &\triangleq \hat{\mathbf{F}}_k^\top \nabla^2 \mathcal{V}(z_k) \hat{\mathbf{G}}_k. \end{aligned} \quad (13)$$

Eq. (13) is pathwise: it uses the realized oracle pair $(\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k)$ rather than the population directions $(\mathbf{F}(z_k), \mathbf{G}(z_k))$. Define its reference quadratic once as

$$q_k(\beta, \gamma) \triangleq -\beta e_k - \gamma d_k + \frac{1}{2} c_k \beta^2 - b_k \beta \gamma + \frac{1}{2} a_k \gamma^2. \quad (14)$$

Thus, Eq. (12) is $\mathcal{J}_k(\beta, \gamma) = q_k(\beta, \gamma) + \text{rem}_k(\beta, \gamma)$. Here e_k, d_k are first-order predicted decreases along $-\hat{\mathbf{F}}_k, +\hat{\mathbf{G}}_k$, c_k, a_k are their directional curvatures, and b_k is the cross-curvature. Assumptions 3 and 2 respectively control the third-order remainder and the conditional oracle errors used in the expected-drift analysis.

C. Closed-Form Step-Size Selection

Discarding the bounded third-order remainder in Eq. (12) yields a bivariate quadratic model for selecting (β, γ) . Eq. (13) defines reference coefficients from $\nabla \mathcal{V}(z_k)$ and directional Hessian actions. We denote their evaluated counterparts returned by the local-model routine by $(\tilde{e}_k, \tilde{d}_k, \tilde{c}_k, \tilde{a}_k, \tilde{b}_k)$. Exact automatic

differentiation gives $(\tilde{e}_k, \tilde{d}_k, \tilde{c}_k, \tilde{a}_k, \tilde{b}_k) = (e_k, d_k, c_k, a_k, b_k)$, whereas finite-difference or directional-secant implementations, possibly combined with moving averages, generally give estimates such as $\tilde{e}_k \approx e_k$ and $\tilde{c}_k \approx c_k$. The quadratic coefficient matrix assembled from the evaluated values is $\widetilde{\mathbf{H}}_k \triangleq \begin{bmatrix} \tilde{c}_k & -\tilde{b}_k \\ -\tilde{b}_k & \tilde{a}_k \end{bmatrix}$. Because local nonconvexity or evaluation error may make $\widetilde{\mathbf{H}}_k$ indefinite or nearly singular, let $\lambda_{\min}(\cdot)$ denote the smallest eigenvalue, fix a positive-definiteness floor $\lambda_{\text{pd}} > 0$, define $[u]_+ \triangleq \max\{u, 0\}$, and apply the common spectral shift

$$\lambda_k = [\lambda_{\text{pd}} - \lambda_{\min}(\widetilde{\mathbf{H}}_k)]_+, \quad \widehat{\mathbf{H}}_k = \widetilde{\mathbf{H}}_k + \lambda_k \mathbf{I}_2. \quad (15)$$

Here λ_k is the smallest nonnegative common diagonal correction that raises the minimum eigenvalue to at least λ_{pd} . We denote the resulting numerically stabilized coefficients by hats: $\hat{e}_k = \tilde{e}_k$, $\hat{d}_k = \tilde{d}_k$, $\hat{b}_k = \tilde{b}_k$, and $(\hat{c}_k, \hat{a}_k) = (\tilde{c}_k + \lambda_k, \tilde{a}_k + \lambda_k)$. Equivalently, $\rho_{c,k} = \rho_{a,k} = \lambda_k \geq 0$ add $\frac{1}{2}\rho_{c,k}\beta^2 + \frac{1}{2}\rho_{a,k}\gamma^2$ to the tilde-coefficient model while leaving its linear and cross terms unchanged. The implemented model is therefore

$$\hat{q}_k(\beta, \gamma) \triangleq -\beta\hat{e}_k - \gamma\hat{d}_k + \frac{1}{2}\hat{c}_k\beta^2 - \hat{b}_k\beta\gamma + \frac{1}{2}\hat{a}_k\gamma^2, \quad (\beta, \gamma) \in [0, \infty)^2. \quad (16)$$

Its stabilized Hessian is $\widehat{\mathbf{H}}_k$, and $\hat{\delta}_k \triangleq \det \widehat{\mathbf{H}}_k = \hat{c}_k\hat{a}_k - \hat{b}_k^2$.

Eq. (15) gives $\lambda_{\min}(\widehat{\mathbf{H}}_k) = \max\{\lambda_{\min}(\widetilde{\mathbf{H}}_k), \lambda_{\text{pd}}\} \geq \lambda_{\text{pd}}$. Thus, $\widehat{\mathbf{H}}_k \succ 0$, and its leading principal minors satisfy $\hat{c}_k > 0$ and $\hat{\delta}_k > 0$. The controller thus solves the strictly convex two-variable quadratic program $\min_{(\beta, \gamma) \in [0, \infty)^2} \hat{q}_k(\beta, \gamma)$. Its 2×2 Hessian permits an explicit interior solution together with a finite set of boundary candidates. The unconstrained critical point is

$$(\beta_k^{\text{uc}}, \gamma_k^{\text{uc}}) \triangleq \left(\frac{\hat{a}_k\hat{e}_k + \hat{b}_k\hat{d}_k}{\hat{\delta}_k}, \frac{\hat{b}_k\hat{e}_k + \hat{c}_k\hat{d}_k}{\hat{\delta}_k} \right), \quad (17)$$

and the boundary candidates are $([\hat{e}_k/\hat{c}_k]_+, 0)$, $(0, [\hat{d}_k/\hat{a}_k]_+)$, $(0, 0)$.

Proposition 2 (Closed-form two-direction surrogate controller). *The implemented problem in Eq. (16) admits a unique minimizer, given by Eq. (17) when both components are nonnegative, and otherwise by the boundary candidate with smallest $\hat{q}_k$. When the unconstrained critical point belongs to $[0, \infty)^2$, the implemented predicted decrease $\widehat{\Delta}_k^* \triangleq -\hat{q}_k(\beta_k^*, \gamma_k^*)$ is*

$$\widehat{\Delta}_k^* = \frac{\hat{a}_k\hat{e}_k^2 + 2\hat{b}_k\hat{e}_k\hat{d}_k + \hat{c}_k\hat{d}_k^2}{2\hat{\delta}_k}. \quad (18)$$

This gives the uncapped closed form; the practical step-size safeguards require the exact box solution below.

Proposition 3 (Exact box-constrained fallback). *Fix finite caps $\beta_{\text{max}}, \gamma_{\text{max}} > 0$. The unique minimizer of $\hat{q}_k$ on $[0, \beta_{\text{max}}] \times [0, \gamma_{\text{max}}]$ is the member with smallest objective value among the feasible unconstrained point in Eq. (17) and the four edge minimizers*

$$\begin{aligned} &([\hat{e}_k/\hat{c}_k]_{[0, \beta_{\text{max}}]}, 0), \quad (0, [\hat{d}_k/\hat{a}_k]_{[0, \gamma_{\text{max}}]}), \\ &(\beta_{\text{max}}, [\hat{d}_k + \hat{b}_k\beta_{\text{max}}]/\hat{a}_k)_{[0, \gamma_{\text{max}}]}), \\ &([\hat{e}_k + \hat{b}_k\gamma_{\text{max}}]/\hat{c}_k)_{[0, \beta_{\text{max}}]}, \gamma_{\text{max}}), \end{aligned}$$

where $[u]_{[l, r]} \triangleq \min\{r, \max\{l, u\}\}$. Consequently, its predicted decrease is at least as large as that of every one-direction or fixed-coupling comparator whose step-size pair lies in the same box.

Proof. Strict convexity gives a unique minimizer. If it is in the interior, the gradient equations give Eq. (17); otherwise it lies on one of the four edges, and differentiating the resulting one-dimensional strictly convex quadratic and clipping its stationary point to that edge gives the four displayed candidates. Set inclusion then proves the stated comparison. $\square$

D. Model-Level Curvature Improvement

The closed-form rule makes the two-direction update computationally explicit. Its model-level benefit is determined by comparing the optimized quadratic decrease with the decreases attainable after restricting the feasible step-size pair.

Proposition 4 (Predicted-decrease comparison). *Fix $s_{\text{max}} > 0$ and define*

$$\begin{aligned} \widehat{\Delta}_k^{\mathbf{F}} &\triangleq -\min_{\beta \geq 0} \hat{q}_k(\beta, 0) = \frac{[\hat{e}_k]_+^2}{2\hat{c}_k}, \\ \widehat{\Delta}_k^{\mathbf{G}} &\triangleq -\min_{\gamma \geq 0} \hat{q}_k(0, \gamma) = \frac{[\hat{d}_k]_+^2}{2\hat{a}_k}, \\ \widehat{\Delta}_k^{2\text{nd}} &\triangleq -\min_{s \in [0, s_{\text{max}}]} \hat{q}_k(s, s^2), \end{aligned}$$

Here $\widehat{\Delta}_k^{2\text{nd}}$ is the PPM/EGM-style second-order coupling comparator $(\beta, \gamma) = (s, s^2)$. Then $\widehat{\Delta}_k^* \geq \max\{\widehat{\Delta}_k^{\mathbf{F}}, \widehat{\Delta}_k^{\mathbf{G}}, \widehat{\Delta}_k^{2\text{nd}}\}$ almost surely. Moreover, when the unconstrained critical point in Eq. (17) belongs to $[0, \infty)^2$, the gap over the pure-field rule is exactly

$$\widehat{\Delta}_k^* - \widehat{\Delta}_k^{\mathbf{F}} = \frac{(\hat{b}_k\hat{e}_k + \hat{c}_k\hat{d}_k)^2}{2\hat{c}_k\hat{\delta}_k} + \frac{[-\hat{e}_k]_+^2}{2\hat{c}_k} \geq 0, \quad (19)$$

which is strictly positive if and only if either $\hat{b}_k\hat{e}_k + \hat{c}_k\hat{d}_k \neq 0$ or $\hat{e}_k < 0$. In the usual field-descent regime $\hat{e}_k \geq 0$, strict improvement is therefore equivalent to the curvature direction having a nonzero reduced gradient after projecting out the field direction in the $\widehat{\mathbf{H}}_k$ -metric.

Eq. (19) measures the curvature step's extra predicted decrease through the reduced gradient $\hat{b}_k\hat{e}_k + \hat{c}_k\hat{d}_k$ left after accounting for the field direction. For collinear directions, the common spectral shift makes the Hessian nonsingular, but any gain then comes from step-size constraints or regularization because the direction span is unchanged. Section IV combines this comparator dominance with the field-geometry certificate and explicitly propagates stochastic-oracle, model-estimation, Taylor-remainder, and safeguard errors into finite-time guarantees.

IV. FINITE-TIME STATIONARITY ANALYSIS

Section III constructs the controller and characterizes its predicted quadratic decrease. We now bound its realized and conditional Lyapunov changes, combine geometric coverage with box-QP dominance for finite-time stationarity, and then transfer stationarity to policy-space performance.

A. From Field Geometry to Conditional Drift

Assumption 4 (Local admissible region). *There exist an open set $\mathcal{N}$ whose compact closure satisfies $\bar{\mathcal{N}} \subset \mathcal{D}$ and an index $k_0 \geq 0$ such that $\mathcal{N}$ contains the local stationary component of interest, $\mathcal{P}$ is C^3 on a neighborhood of $\bar{\mathcal{N}}$, and $\mathbf{z}_k \in \mathcal{N}$ for every $k \geq k_0$ almost surely.*

This is a local-invariance condition: the safeguards in Section V keep the accepted trajectory in a prescribed implementable region, and the analysis applies when that trajectory remains in $\mathcal{N}$. On this region, define the geometric coverage modulus

$$m_{\text{geo}} \triangleq \inf_{\substack{\mathbf{z} \in \mathcal{N} \\ \Phi(\mathbf{z}) > 0}} \frac{\max\{\chi_F(\mathbf{z}), \chi_G(\mathbf{z})\}}{\Phi(\mathbf{z})}. \quad (20)$$

The stationarity result below applies when $m_{\text{geo}} > 0$, meaning that at every nonstationary point at least one available coordinate supplies a uniformly proportional field-energy decrease. Unlike a gradient-aligned field, a general saddle field does not make the positivity in Eq. (20) automatic; the modulus is the quantitative step that converts directional field-energy decrease into the stationarity residual Φ .

The following condition derives neighborhood coverage from the linearization at the target stationary point; $\sigma_{\min}(\cdot)$ denotes the smallest singular value.

Proposition 5 (A sufficient condition for geometric coverage). *Fix a target stationary point $\mathbf{z}^* \in \mathcal{Z}^*$ and define its metric-transformed Jacobian $\tilde{\mathbf{A}}_* \triangleq \mathbf{W}_V^{1/2} \mathbf{A}(\mathbf{z}^*) \mathbf{W}_V^{-1/2}$. If $\text{sym}(\tilde{\mathbf{A}}_*) \succeq 0$ and $\sigma_{\min}(\tilde{\mathbf{A}}_*) > 0$, then there exist a constant $c_* > 0$ and an open neighborhood $\mathcal{N}_*$ of $\mathbf{z}^*$, with $\bar{\mathcal{N}}_* \subset \mathcal{D}$, such that $\max\{\chi_F(\mathbf{z}), \chi_G(\mathbf{z})\} \geq c_* \Phi(\mathbf{z})$ for every $\mathbf{z} \in \mathcal{N}_*$. Consequently, if the admissible region in Assumption 4 is chosen with $\bar{\mathcal{N}} \subset \mathcal{N}_*$, then $m_{\text{geo}} \geq c_* > 0$.*

For $\mathbf{W}_V = \mathbf{I}$, max-variable concavity and min-variable convexity at $\mathbf{z}^*$ give the positive-semidefinite condition, while nonsingularity gives a nondegenerate target linearization; continuity then supplies neighborhood coverage, including pure rotation with $\text{sym}(\tilde{\mathbf{A}}_*) = 0$. For the normal block $\mathbf{A} = \mu \mathbf{I}_2 + \sigma J_0$, the exact ratio in Eq. (20) is $\max\{\mu, \sigma^2 - \mu^2\}$; in particular, $\mu = 0$ gives $m_{\text{geo}} = \sigma^2$.

Compactness and smoothness give the finite performance-direction bound

$$B_{P,D} \triangleq \sup_{\mathbf{z} \in \bar{\mathcal{N}}} \max\{|\langle \nabla \mathcal{P}, \mathbf{F} \rangle|, |\langle \nabla \mathcal{P}, \mathbf{G} \rangle|\}.$$

Lemma 4 (Geometry-to-drift certificate). *Under Assumption 4, suppose the modulus in Eq. (20) satisfies $m_{\text{geo}} > 0$. Define the geometric descent coefficient μ_{geo} and policy-component perturbation r_P by*

$$\mu_{\text{geo}} \triangleq \frac{\lambda_F m_{\text{geo}}}{\kappa_F}, \quad r_P \triangleq \frac{\lambda_P B_{P,D}}{\kappa_P}.$$

Then, for every $\mathbf{z} \in \mathcal{N}$,

$$\max\{D_F(\mathbf{z}), D_G(\mathbf{z})\} \geq \mu_{\text{geo}} \Phi(\mathbf{z}) - r_P. \quad (21)$$

Proposition 5 and Lemma 4 form the first bridge: target-Jacobian conditions produce neighborhood coverage, which converts one of the two field-energy signals into the composite decrease in Eq. (21) without directly assuming uniform

alignment with $-\nabla \mathcal{V}$. The policy component contributes only the finite perturbation r_P , which vanishes for $\lambda_P = 0$ and otherwise enters the residual floor.

Because $\bar{\mathcal{N}}$ is compactly contained in $\mathcal{D}$, continuity yields uniform constants $G_V, H_V, B_F, B_G \geq 0$ such that $\|\nabla \mathcal{V}\| \leq G_V$, $\|\nabla^2 \mathcal{V}\| \leq H_V$, $\|\mathbf{F}\| \leq B_F$, and $\|\mathbf{G}\| \leq B_G$ on $\mathcal{N}$. In particular, $G_{V,k}, H_{V,k}$ may be replaced by G_V, H_V for $k \geq k_0$.

Recall the pathwise reference quadratic q_k in Eq. (14) and the stabilized implemented quadratic $\hat{q}_k$ in Eq. (16). For fixed caps $\beta_{\max}, \gamma_{\max} > 0$ and an implemented pair $(\beta_k, \gamma_k) \in [0, \beta_{\max}] \times [0, \gamma_{\max}]$, define the box-optimal predicted decrease

$$\hat{\Delta}_k^{\text{box}} \triangleq - \min_{0 \leq \beta \leq \beta_{\max}, 0 \leq \gamma \leq \gamma_{\max}} \hat{q}_k(\beta, \gamma).$$

The implemented model $\hat{q}_k$ uses the stabilized coefficients, and one-sided residuals measure model discrepancy and accepted-step suboptimality. Define the cubic Taylor remainder $\varepsilon_k^{\text{rem}} \triangleq M_{3,k}(\beta_k \|\hat{\mathbf{F}}_k\| + \gamma_k \|\hat{\mathbf{G}}_k\|)^3/6$. Let $\varepsilon_k^{\text{model}} \geq 0$ be $\mathcal{F}_{k+1}$ -measurable and satisfy $q_k(\beta_k, \gamma_k) \leq \hat{q}_k(\beta_k, \gamma_k) + \varepsilon_k^{\text{model}}$. For the implemented box problem, define $\varepsilon_k^{\text{step,box}} \triangleq \hat{q}_k(\beta_k, \gamma_k) + \hat{\Delta}_k^{\text{box}} \geq 0$ and

$$\varepsilon_k^{\text{box}} \triangleq \varepsilon_k^{\text{rem}} + \varepsilon_k^{\text{model}} + \varepsilon_k^{\text{step,box}}. \quad (22)$$

Proposition 6 (One-sided local-model perturbation bounds). *Suppose the raw evaluated coefficients satisfy $|\tilde{e}_k - e_k| \leq \delta_{e,k}$, $|\tilde{d}_k - d_k| \leq \delta_{d,k}$, $|\tilde{c}_k - c_k| \leq \delta_{c,k}$, $|\tilde{a}_k - a_k| \leq \delta_{a,k}$, and $|\tilde{b}_k - b_k| \leq \delta_{b,k}$ for nonnegative error bounds. Then, for every feasible (β, γ) ,*

$$\begin{aligned} \varepsilon_k^{\text{model}}(\beta, \gamma) &\triangleq \beta \delta_{e,k} + \gamma \delta_{d,k} + \frac{1}{2} \beta^2 \delta_{c,k} + \beta \gamma \delta_{b,k} + \frac{1}{2} \gamma^2 \delta_{a,k}, \\ \varepsilon_k^{\text{cmp}}(\beta, \gamma) &\triangleq \varepsilon_k^{\text{model}}(\beta, \gamma) + \frac{1}{2} \rho_{c,k} \beta^2 + \frac{1}{2} \rho_{a,k} \gamma^2, \\ q_k - \hat{q}_k &\leq \varepsilon_k^{\text{model}}, \quad \hat{q}_k - q_k \leq \varepsilon_k^{\text{cmp}}. \end{aligned} \quad (23)$$

In the last line, all arguments (β, γ) are suppressed, and the superscript cmp denotes the comparator-side bound. In particular, the one-sided relation used in Eq. (22) holds with $\varepsilon_k^{\text{model}} = \varepsilon_k^{\text{model}}(\beta_k, \gamma_k)$.

Exact automatic differentiation makes the raw coefficient errors and the selected-step residual $\varepsilon_k^{\text{model}}$ zero because the spectral shift only raises $\hat{q}_k$; the comparator-side bound retains its explicit shift cost, and an exact box-QP solution makes the box-step residual zero.

Lemma 5 (Pathwise one-step Lyapunov bound). *Under Assumption 3, for every k ,*

$$\mathcal{V}(\mathbf{z}_{k+1}) - \mathcal{V}(\mathbf{z}_k) \leq -\hat{\Delta}_k^{\text{box}} + \varepsilon_k^{\text{box}} \quad \text{almost surely}. \quad (24)$$

This inequality separates the decrease predicted by the QP from stochastic, modeling, and safeguard residuals.

Lemma 6 (Conditional Taylor-remainder bound). *Suppose Assumptions 2 and 3 hold and $\beta_k \leq \beta_{\max}, \gamma_k \leq \gamma_{\max}$. Then*

$$\mathbb{E}[\varepsilon_k^{\text{rem}} \mid \mathcal{F}_k] \leq \frac{2M_{3,k}}{3} (\beta_{\max} + \gamma_{\max})^3 \sigma_{3,k}^3. \quad (25)$$

If $M_{3,k} \leq \bar{M}_3$ and $\sigma_{3,k} \leq \bar{\sigma}_3$, then $\mathbb{E}[\varepsilon_k^{\text{rem}} \mid \mathcal{F}_k] \leq \bar{R}_{\text{rem}}$ almost surely, where $\bar{R}_{\text{rem}} \triangleq (2\bar{M}_3/3)(\beta_{\max} + \gamma_{\max})^3 \bar{\sigma}_3^3$.

Assumption 5 (Uniform local budgets). *There exist finite nonnegative constants appearing below such that, for every $k \geq k_0$, the following hold: (a) $\bar{b}_k^F \leq \bar{b}_F, \bar{b}_k^G \leq \bar{b}_G, \sigma_{2,k} \leq$*

$\bar{\sigma}_2$, and $\sigma_{3,k} \leq \bar{\sigma}_3$; (b) the common spectral shift satisfies $\rho_{c,k} = \rho_{a,k} \leq \bar{\rho}$, and the raw coefficient-evaluation errors in Proposition 6 satisfy

$$\begin{aligned} \mathbb{E}[\delta_{e,k} | \mathcal{F}_k] &\leq \bar{\delta}_e, \quad \mathbb{E}[\delta_{d,k} | \mathcal{F}_k] \leq \bar{\delta}_d, \\ \mathbb{E}[\delta_{c,k} | \mathcal{F}_k] &\leq \bar{\delta}_c, \quad \mathbb{E}[\delta_{a,k} | \mathcal{F}_k] \leq \bar{\delta}_a, \quad \mathbb{E}[\delta_{b,k} | \mathcal{F}_k] \leq \bar{\delta}_b; \\ (c) \quad M_{3,k} &\leq \bar{M}_3 \text{ and } \mathbb{E}[\varepsilon_k^{\text{step,box}} | \mathcal{F}_k] \leq \bar{R}_{\text{step}} \text{ almost surely.} \end{aligned}$$

Define the derived oracle-moment bounds and aggregate residual budgets

$$\begin{aligned} M_F &\triangleq 3B_F^2 + 3\bar{b}_F^2 + 3\bar{\sigma}_2^2, \quad M_G \triangleq 3B_G^2 + 3\bar{b}_G^2 + 3\bar{\sigma}_2^2, \\ \bar{b}_D &\triangleq \max\{\bar{b}_F, \bar{b}_G\}, \quad M_D \triangleq \max\{M_F, M_G\}, \\ \bar{\delta}_1 &\triangleq \max\{\bar{\delta}_e, \bar{\delta}_d\}, \quad \bar{\delta}_2 \triangleq \max\{\bar{\delta}_c, \bar{\delta}_a\}, \\ \bar{R}_{\text{model}} &\triangleq \beta_{\max} \bar{\delta}_e + \gamma_{\max} \bar{\delta}_d + \frac{1}{2} \beta_{\max}^2 \bar{\delta}_c \\ &\quad + \beta_{\max} \gamma_{\max} \bar{\delta}_b + \frac{1}{2} \gamma_{\max}^2 \bar{\delta}_a, \\ \bar{R}_{\text{box}} &\triangleq \bar{R}_{\text{rem}} + \bar{R}_{\text{model}} + \bar{R}_{\text{step}}. \end{aligned}$$

Lemma 6, Proposition 6, the caps, and Assumption 5 give $\mathbb{E}[\varepsilon_k^{\text{box}} | \mathcal{F}_k] \leq \bar{R}_{\text{box}}$ almost surely, so the aggregate is derived from its three displayed sources. For a comparator scale $s \in [0, \min\{\beta_{\max}, \gamma_{\max}\}]$, let

$$\mathcal{R}_D(s) \triangleq s(G_V \bar{b}_D + \bar{\delta}_1) + \frac{1}{2} s^2 (H_V M_D + \bar{\rho} + \bar{\delta}_2) + \bar{R}_{\text{box}}. \quad (26)$$

Lemma 7 (Conditional Lyapunov-drift comparator bound). *Suppose Assumptions 1, 2, 3, 4, and 5 hold. Let $j_k \in \{F, G\}$ be any $\mathcal{F}_k$ -measurable selector and write $D_{j_k} = D_F$ or D_G accordingly. Then, for every admissible deterministic comparator scale s ,*

$$\mathcal{D}_k \leq -s D_{j_k}(\mathbf{z}_k) + \mathcal{R}_D(s). \quad (27)$$

Thus, the implemented update inherits, up to the explicit residual budget, the conditional drift decrease certified by either pure coordinate.

The selector is used only in the proof; the implemented box QP optimizes over both pure-coordinate comparators and their combinations. This second bridge uses QP optimality and the four explicit residual budgets to lift a comparison on each realized stochastic stencil to the conditional-drift bound in Eq. (27).

B. Finite-Time Stationarity Guarantee

Theorem 2 (Main Theorem: Geometry-Adaptive Finite-Time Stationarity). *Suppose Assumptions 1, 2, 3, 4, and 5 hold, and the modulus in Eq. (20) satisfies $m_{\text{geo}} > 0$. For any comparator scale $s \in (0, \min\{\beta_{\max}, \gamma_{\max}\}]$, define $\mathcal{R}_{\text{geo}}(s) \triangleq \mathcal{R}_D(s) + sr_P$. Then, for every integer $K > k_0$,*

$$\frac{1}{K - k_0} \sum_{k=k_0}^{K-1} \mathbb{E}[\Phi(\mathbf{z}_k)] \leq \frac{\mathbb{E}[\mathcal{V}(\mathbf{z}_{k_0})]}{s m_{\text{geo}}(K - k_0)} + \frac{\mathcal{R}_{\text{geo}}(s)}{s m_{\text{geo}}}. \quad (28)$$

For fixed admissible s and K -independent budgets, Eq. (28) gives an $O(1/K)$ transient to an explicit floor containing the oracle, model, Taylor-remainder, policy-component, and safeguard errors. The signals χ_F and χ_G respectively certify contractive symmetric fields and pure rotation, and box-QP dominance inherits either certificate. For an independent uniform index $I_K \in \{k_0, \dots, K-1\}$, the left side is $\mathbb{E}[\Phi(\mathbf{z}_{I_K})]$, so no parameter averaging is required. Increasing the analysis

comparator s strengthens nominal decrease but enlarges higher-order terms, whereas decreasing it magnifies any s -independent residual after normalization; the implemented controller itself still selects (β_k, γ_k) from the box QP. The result is therefore local and ergodic, not a claim that curvature always descends or that convergence is global.

C. Policy-Space Performance Guarantee

Exploitability and robust return are policy-space quantities beyond parameter-space stationarity. We therefore transfer Lyapunov control to these performance measures. For stationary randomized policies, define the maximizing player's worst-case return, game value, and unregularized Nash gap (reported below as exploitability), where the subscript BR denotes best response:

$$\begin{aligned} R_{\text{BR}}(\pi) &\triangleq \inf_{\bar{\nu}} J(\pi, \bar{\nu}), \\ v^* &\triangleq \sup_{\bar{\pi}} R_{\text{BR}}(\bar{\pi}), \\ \text{Gap}_0(\pi, \nu) &\triangleq \sup_{\bar{\pi}} J(\bar{\pi}, \nu) - \inf_{\bar{\nu}} J(\pi, \bar{\nu}). \end{aligned} \quad (29)$$

The following proposition links the entropy-regularized gap used by the controller to the original zero-sum Markov-game objective.

Proposition 7 (Policy-space performance bridge). *Suppose $\mathcal{A}$ and $\mathcal{B}$ are finite. For every policy pair (π, ν) and $\varepsilon \geq 0$,*

$$\begin{aligned} 0 &\leq v^* - R_{\text{BR}}(\pi) \leq \text{Gap}_0(\pi, \nu) \\ &\leq \text{Gap}_{\varepsilon}(\pi, \nu) + \frac{\varepsilon(\log |\mathcal{A}| + \log |\mathcal{B}|)}{1 - \alpha}. \end{aligned} \quad (30)$$

Proof. The definitions in Eq. (29) give the first inequality, while $v^* \leq \sup_{\bar{\pi}} J(\bar{\pi}, \nu)$ gives the second. Write $A_{\alpha} = \log |\mathcal{A}|/(1 - \alpha)$ and $B_{\alpha} = \log |\mathcal{B}|/(1 - \alpha)$. The per-state entropy bound gives $0 \leq H_{\alpha}^{\pi} \leq A_{\alpha}$ and $0 \leq H_{\alpha}^{\nu} \leq B_{\alpha}$ for every policy pair. These bounds imply

$$\begin{aligned} \sup_{\bar{\pi}} J(\bar{\pi}, \nu) &\leq \sup_{\bar{\pi}} J_{\varepsilon}(\bar{\pi}, \nu) + \varepsilon B_{\alpha}, \\ -\inf_{\bar{\nu}} J(\pi, \bar{\nu}) &\leq -\inf_{\bar{\nu}} J_{\varepsilon}(\pi, \bar{\nu}) + \varepsilon A_{\alpha}. \end{aligned}$$

Adding the last two inequalities gives Eq. (30). $\square$

Thus the regularized gap controls the maximizing player's unregularized worst-case-return loss up to the stated entropy term.

Proposition 8 (Smooth finite-game performance component). *Suppose $\mathcal{S}, \mathcal{A}, \mathcal{B}$ are finite, the one-stage rewards are bounded, $\varepsilon > 0$, and, on the parameter neighborhood under consideration, the current policies are C^3 with action probabilities bounded away from zero. Then the entropy-regularized best-response values against either current player, and hence Gap_{ε} , are C^3 functions of the current policy parameters.*

Proof. Fix the minimizing-player policy. The maximizing player's soft Bellman value satisfies $v = T_{\varepsilon}(v; \psi)$, where each component of T_{ε} is an $\varepsilon \log \sum \exp$ of the action values induced by ν_{ψ} . Explicitly,

$$[T_{\varepsilon}(v; \psi)]_s = \varepsilon \log \sum_a \exp \left(\frac{Q_{\psi}^v(s, a)}{\varepsilon} \right),$$

$$Q_\psi^v(s, a) \triangleq \bar{r}_\psi(s, a) + \alpha \sum_{s'} \bar{P}_\psi(s'|s, a) v_{s'}.$$

Here $\mathcal{H}(p) \triangleq -\sum_i p_i \log p_i$ is the per-state Shannon entropy, $\bar{P}_\psi$ averages P over ν_ψ , and $\bar{r}_\psi$ averages r while including the opponent-entropy term $-\varepsilon \mathcal{H}(\nu_\psi(\cdot|s))$ from Eq. (1). The map is C^3 in (v, ψ) , and its Jacobian with respect to v is $\alpha P_{\pi_{\text{BR}}, \nu_\psi}$, where π_{BR} is the maximizing player's regularized best response and $P_{\pi_{\text{BR}}, \nu_\psi}$ is the induced transition matrix. Thus, with D_v denoting differentiation in v , $\|D_v T_\varepsilon\|_\infty \leq \alpha < 1$. Hence $I - D_v T_\varepsilon$ is invertible by the Neumann series, and the implicit-function theorem makes the unique soft fixed point C^3 in ψ . The minimizing player's soft-min Bellman equation follows after replacing $\log \sum \exp$ by $-\log \sum \exp(-\cdot)$. Multiplication by ρ gives the two regularized best-response values, whose difference is Gap_ε . $\square$

Consequently, choosing $\mathcal{P} = \text{Gap}_\varepsilon$ gives a concrete smooth performance component for Definition 1 in finite Markov games.

Assumption 6 (Local policy-space error bound). *There exist constants $c_{\text{gap}} > 0$ and $\xi_{\text{gap}} \geq 0$ such that*

$$\text{Gap}_\varepsilon(\mathbf{z}) \leq c_{\text{gap}} \Phi(\mathbf{z}) + \xi_{\text{gap}}, \quad \mathbf{z} \in \mathcal{N}. \quad (31)$$

Eq. (31) is a local Polyak–Łojasiewicz-type policy error bound [26]. The slack ξ_{gap} permits parameter-stationary points with nonzero policy-space exploitability, so Eq. (31) does not require the two zero sets to coincide.

Corollary 1 (Ergodic policy-space performance). *Suppose the assumptions of Theorem 2 and Assumption 6 hold, and $\mathcal{A}$ and $\mathcal{B}$ are finite. Then*

$$\begin{aligned} & \frac{1}{K - k_0} \sum_{k=k_0}^{K-1} \mathbb{E}[v^* - R_{\text{BR}}(\pi_{\theta_k})] \\ & \leq \frac{1}{K - k_0} \sum_{k=k_0}^{K-1} \mathbb{E}[\text{Gap}_0(\pi_{\theta_k}, \nu_{\psi_k})] \\ & \leq c_{\text{gap}} \left[\frac{\mathbb{E}[\mathcal{V}(\mathbf{z}_{k_0})]}{s\mu_{\text{geo}}(K - k_0)} + \frac{\mathcal{R}_{\text{geo}}(s)}{s\mu_{\text{geo}}} \right] + \xi_{\text{gap}} + B_\varepsilon, \end{aligned} \quad (32)$$

where $B_\varepsilon \triangleq \varepsilon(\log |\mathcal{A}| + \log |\mathcal{B}|)/(1 - \alpha)$ is the entropy-regularization bias bound.

Proof. Average Eq. (31), apply Theorem 2, and then use Proposition 7 pointwise. $\square$

Thus, Eq. (32) gives the policy-space consequence of the main stationarity theorem under the local error bound with slack.

V. PRACTICAL IMPLEMENTATION

This section specifies how the local quantities required by the controller can be evaluated in practice. We consider exact automatic differentiation, directional-secant approximations, and safeguarded box-constrained updates.

A. Exact local-model evaluation

The reference pathwise terms in Eq. (13) are assembled from one field oracle $\hat{\mathbf{F}}_k$, one curvature oracle $\hat{\mathbf{G}}_k$, one gradient $\nabla \mathcal{V}(\mathbf{z}_k)$, and the Hessian–vector products of $\mathcal{V}$ along $\hat{\mathbf{F}}_k$ and $\hat{\mathbf{G}}_k$. Automatic differentiation can pass through a fixed differentiable inner loop for $\mathcal{P}$, while implicit differentiation handles an exact local envelope. Because $\mathcal{E}_F$ contains $\mathbf{F}^\top \mathbf{W}_V \mathbf{F}/2$, exact Hessian–vector products of $\mathcal{V}$ generally involve third derivatives of the game objective; the secant realization below avoids forming them. Exact automatic differentiation gives $(\tilde{e}_k, \tilde{d}_k, \tilde{c}_k, \tilde{a}_k, \tilde{b}_k) = (e_k, d_k, c_k, a_k, b_k)$. After exact coefficient evaluation, $\hat{q}_k - q_k = \lambda_k(\beta^2 + \gamma^2)/2 \geq 0$; hence the selected-step one-sided model residual is zero, while the comparator-side bound retains the explicit shift.

B. Approximate local-model estimation

For probe radii $\delta_{F,k}, \delta_{G,k} > 0$, a directional-secant realization holds $(\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k)$ fixed and evaluates $V_{\pm F} = \mathcal{V}(\mathbf{z}_k \pm \delta_{F,k} \hat{\mathbf{F}}_k)$, $V_{\pm G} = \mathcal{V}(\mathbf{z}_k \pm \delta_{G,k} \hat{\mathbf{G}}_k)$, and $V_{\pm F, \pm G} = \mathcal{V}(\mathbf{z}_k \pm \delta_{F,k} \hat{\mathbf{F}}_k \pm \delta_{G,k} \hat{\mathbf{G}}_k)$. Writing $V_0 = \mathcal{V}(\mathbf{z}_k)$, the central stencils give the tilde approximations

$$\begin{aligned} \tilde{e}_k &= \frac{V_{+F} - V_{-F}}{2\delta_{F,k}}, & \tilde{d}_k &= -\frac{V_{+G} - V_{-G}}{2\delta_{G,k}}, \\ \tilde{c}_k &= \frac{V_{+F} - 2V_0 + V_{-F}}{\delta_{F,k}^2}, & \tilde{a}_k &= \frac{V_{+G} - 2V_0 + V_{-G}}{\delta_{G,k}^2}, \\ \tilde{b}_k &= \frac{V_{+F,+G} - V_{+F,-G} - V_{-F,+G} + V_{-F,-G}}{4\delta_{F,k}\delta_{G,k}}, \end{aligned}$$

before the spectral shift in Section III-C is applied. Writing $\hat{\mathbf{F}}(\mathbf{z}, \zeta)$ for the field oracle evaluated at $\mathbf{z}$ with random seed ζ , a shared seed ζ_k and radius $\epsilon_k > 0$ give the Jacobian–vector finite-difference curvature oracle $\hat{\mathbf{G}}_k^{\text{fd}} = [\hat{\mathbf{F}}(\mathbf{z}_k + \epsilon_k \hat{\mathbf{F}}_k, \zeta_k) - \hat{\mathbf{F}}(\mathbf{z}_k, \zeta_k)]/\epsilon_k$. A moment-based realization smooths the second-order terms by exponential moving averages while preserving the subproblem.

The effect of these approximation errors on the one-step model is quantified by Proposition 6.

C. Safeguarded computation

With positive caps β_{max} and γ_{max} , the implementation solves $\min_{(\beta, \gamma) \in [0, \beta_{\text{max}}] \times [0, \gamma_{\text{max}}]} \hat{q}_k(\beta, \gamma)$. Any loss from approximate solution or backtracking relative to the box optimum is recorded in $\varepsilon_k^{\text{step, box}}$ in Theorem 2. Each iteration queries $(\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k)$, evaluates the five tilde coefficients, applies the spectral shift, solves the two-dimensional box QP by Proposition 3, and applies Eq. (9). If the trial segment leaves a prescribed implementable trust region $\mathcal{T} \subset \mathcal{D}$, a common backtracking factor shrinks (β_k, γ_k) ; the analysis assumes the accepted trajectory lies in $\mathcal{N}$, and records lost model decrease in $\varepsilon_k^{\text{step, box}}$. For bounded $\mathcal{T}$, the implementation reduces the probe radii as needed until every directional-stencil point lies in $\mathcal{T}$.

VI. EXPERIMENTS

The experiments follow the same hierarchy as the analysis. The normal-form study first isolates the symmetric–antisymmetric mechanism, the tabular and neural population-oracle studies then test policy-space behavior without trajectory noise, and the finite-trajectory study finally replaces

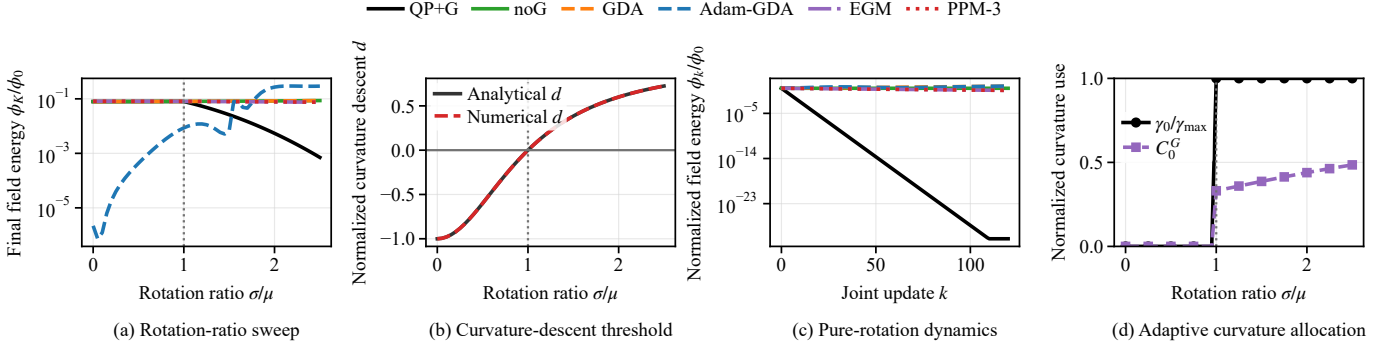


Fig. 2. Controlled normal-form results: (a) final normalized field energy; (b) analytical and computed curvature descent; (c) pure-rotation trajectories; and (d) first-update normalized curvature step and relative curvature contribution. Vertical lines in (a), (b), and (d) mark $\sigma/\mu = 1$.

exact expected directions by sampled stochastic oracles. This progression separates geometric adaptation, nonlinear policy parameterization, and sampling error rather than confounding all three effects in a single benchmark. Methods are the proposed box-QP controller with curvature (QP+G), its field-only ablation noG ($\gamma_k = 0$), GDA, Adam-GDA [27], EGM, and three-inner-step PPM-3. All methods update both players simultaneously; unless noted, fixed steps and QP caps are 0.03, $\eta_\psi = 1$ gives $\mathcal{E}_F = \phi$, and population normalizers are frozen at initialization with floor 10^{-10} . For every Markov-game experiment, $J_{\text{alg}} = J_\varepsilon$ with $(\alpha, \varepsilon) = (0.9, 0.03)$; the regularized and unregularized value tolerances are 10^{-10} and 10^{-12} , and the spectral floor is 10^{-8} . Across matched seeds, all paired 95% confidence intervals are two-sided Student- t intervals for metric-oriented differences, with positive values favoring QP+G.

A. Controlled Normal-Form Geometry

We test the mechanism on the exactly solvable normal linear field with $\lambda_P = 0$, so Eq. (10) retains only field energy. For radial contraction rate $\mu \geq 0$ and angular rotation rate $\sigma \geq 0$,

$$\mathbf{F}(\mathbf{z}) = \mathbf{A}\mathbf{z}, \quad \mathbf{A} = \mu\mathbf{I}_2 + \sigma\mathbf{J}_0, \quad \mathcal{V}(\mathbf{z}) = \frac{1}{2} \|\mathbf{F}(\mathbf{z})\|^2.$$

Here $\mathbf{S} = \mu\mathbf{I}_2$, $\mathbf{W} = \sigma\mathbf{J}_0$, and direct calculation gives

$$\begin{aligned} e &= \langle \nabla \mathcal{V}, \mathbf{F} \rangle = \mu(\mu^2 + \sigma^2) \|\mathbf{z}\|^2, \\ d &= -\langle \nabla \mathcal{V}, \mathbf{G} \rangle = (\mu^2 + \sigma^2)(\sigma^2 - \mu^2) \|\mathbf{z}\|^2. \end{aligned} \quad (33)$$

Eq. (33) shows that $+\mathbf{G}$ decreases the field energy exactly when $|\sigma| > |\mu|$, matching Theorem 1; moreover, $\max\{\chi_F, \chi_G\}/\Phi = \max\{\mu, \sigma^2 - \mu^2\}$ evaluates the coverage modulus in Eq. (20). For this quadratic field-energy function, the two-dimensional Taylor model is exact.

To quantify accepted curvature use, define its relative contribution as $C_k^G \triangleq \gamma_k \|\mathbf{G}(\mathbf{z}_k)\| / \max\{\beta_k \|\mathbf{F}(\mathbf{z}_k)\| + \gamma_k \|\mathbf{G}(\mathbf{z}_k)\|, 10^{-15}\}$. Fig. 2 sweeps $\sigma/\mu \in [0, 2.5]$; panels (a)–(c) recover the threshold and pure-rotation decay, while panel (d) shows first-update $\gamma_0/\gamma_{\max}$ and C_0^G from the common initial point. These quantities measure curvature-step activation and the curvature share of the additive nominal update magnitude; their threshold-like profile follows here from exact directions and $\lambda_P = 0$.

B. Tabular Markov Games with Population Oracles

We span a one-state normal-form benchmark and sequential finite-state games while removing function approximation, using rock-paper-scissors (RPS), five-state CyclicControl, and eight-state FrequencyHopping, each with three actions per player and tabular softmax logits. RPS is a standard multi-agent RL benchmark [28]; Appendix B specifies the games. For both the tabular and neural population experiments, the QP+G/noG controller uses the Markov-game composite Lyapunov function

$$V_{\text{MG}}(\mathbf{z}) = 0.3 \frac{\frac{1}{2} \|\mathbf{F}(\mathbf{z})\|^2}{\kappa_F} + \frac{\text{Gap}_\varepsilon(\mathbf{z})}{\kappa_P}, \quad (34)$$

This corresponds to $\lambda_F = 0.3$ and $\lambda_P = 1$, with both normalizers frozen at initialization. Population automatic differentiation computes $\mathbf{F}$ as the saddle gradient of J_ε and $\mathbf{G} = \nabla \mathbf{F} \mathbf{F}$, each stencil recomputes both regularized BRs, and Eq. (30) holds at every checkpoint.

Across ten initializations and 100 joint updates, QP+G raises worst-case return and lowers exploitability relative to noG in all three games. Exact one-sided paired sign tests of the return gains remain significant after Holm correction across the three games [29]. Fig. 3 shows that the same ordering extends to the regularized gap and field norm, linking the performance gain to improved saddle-field optimization.

C. Neural-Policy Markov Games with Population Oracles

The neural-policy experiment isolates nonlinear parameterization from sampling noise using separate 4–8–3 tanh-softmax policies, forming a 134-dimensional population field on the four games in Appendix B. An exact differentiable value-system solve supplies $\mathbf{F}$ and $\mathbf{G} = \nabla \mathbf{F} \mathbf{F}$; QP+G and noG evaluate Eq. (34) on nine- and three-point central stencils, respectively, with $\delta_D = \min\{10^{-2}, 10^{-3}/\max(\|D\|, 10^{-12})\}$, $D \in \{\mathbf{F}, \mathbf{G}\}$. Five screening seeds select each baseline rate from $\{0.001, 0.003, 0.01, 0.03\}$; ten disjoint test seeds use 60 simultaneous updates. QP+G attains higher worst-case return and lower exploitability than noG in all four games, with positive lower endpoints in all paired 95% confidence intervals for return gain; Fig. 4 displays three games, and SecurityPatrol has the same mean ordering. CyclicControl has the largest exploitability reduction, 79.1%. The game-wise mean of C_k^G ranges from 0.177 to 0.587. With $\mathbf{S}_k \triangleq \text{sym}(\nabla \mathbf{F}(\mathbf{z}_k))$ and $\mathbf{W}_k \triangleq \text{skew}(\nabla \mathbf{F}(\mathbf{z}_k))$, the median ratio

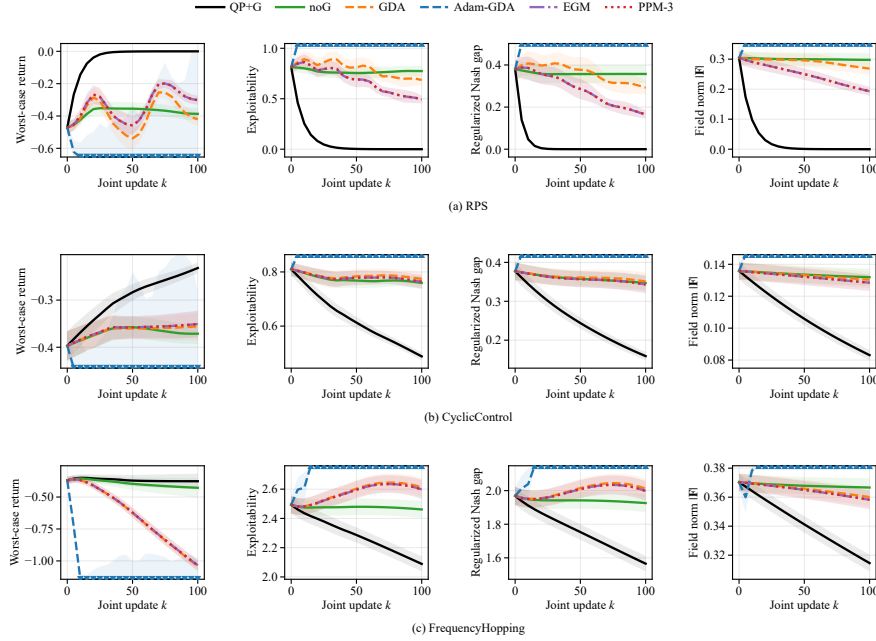


Fig. 3. Tabular population-oracle games over ten seeds (mean $\pm$ standard error); boundary triangles mark Adam-GDA means outside the displayed range.

$\|W_k F(z_k)\| / (\|S_k F(z_k)\| + 10^{-15})$ is 0.951 on Frequency-Hopping and 0.966 on RoutingInterdiction. Thus the controller is not a hard skew-threshold switch: Theorem 1 concerns $+G$ alone, whereas the composite QP may use curvature through the gap gradient and cross-curvature even below the skew threshold.

D. Finite-Trajectory Oracle Validation

The preceding population-oracle studies evaluate the controller with exact expectations and therefore isolate its deterministic geometry. We now test whether the same two-direction mechanism remains observable when the training field, curvature action, and local-model stencil are all estimated from finitely many trajectories. At each of 60 CyclicControl updates, two 4–8–3 policies use 128 horizon-16 trajectories (2048 transitions). One detached-policy batch B_k supplies every stencil through unclipped per-decision prefix ratios and the infinitely differentiable Monte Carlo estimator (DiCE) [30]. DiCE gives the regularized field and Hessian–vector curvature oracles, with the field detached in the latter product. For $\lambda_P = 0$, an independent-batch-normalized field-energy merit controls training, while population quantities are used only for checkpoint evaluation. Proposition 6 and Theorem 2 account for dependent same-batch stencil errors.

TABLE I
FINITE-TRAJECTORY CYCLICCONTROL AT $k = 60$ (2048 TRANSITIONS/UPDATE; TEN HELD-OUT SEEDS). ENTRIES ARE MEAN $\pm$ STANDARD ERROR WITH PAIRED 95% CONFIDENCE INTERVALS; POSITIVE Δ FAVORS QP+G.

Metric	QP+G	noG	Paired Δ
Worst-case return $\uparrow$	-0.203 ± 0.032	-0.546 ± 0.078	$0.342 [0.123, 0.561]$
Exploitability $\downarrow$	0.429 ± 0.047	0.932 ± 0.096	$0.503 [0.274, 0.732]$
Population $\ F\ \downarrow$	0.169 ± 0.025	0.377 ± 0.037	$0.207 [0.134, 0.281]$

All paired intervals are positive; mean exploitability decreases by 54.0% relative to noG, and curvature remains active at every update. The diagnostics separate parameter-space stationarity ($\|F\|$), unilateral policy improvement (exploitability), and unregularized robust performance (worst-case return), while population checkpoints keep training noise out of evaluation. Thus the population-oracle curves test deterministic geometry, whereas Table I tests whether its benefits survive the biased, dependent sampled queries covered by the analysis; matched-seed intervals isolate the curvature-enabled controller.

VII. CONCLUSION

This paper developed complementary field and curvature directions for zero-sum Markov-game policy optimization. The symmetric–antisymmetric analysis identifies when curvature decreases field energy, while a closed-form box QP selects independent step sizes from a second-order Lyapunov-drift model. Under local geometric coverage and explicit stochastic-error budgets, the stationarity guarantee gives an $O(1/K)$ ergodic transient plus an error floor; with the local policy-space error bound, it transfers to Nash-gap and worst-case-return bounds. Population-oracle and finite-trajectory experiments support the predicted geometry and show improved worst-case return and exploitability over the field-only ablation. Independent step sizes thereby use curvature when its local contribution is beneficial without requiring it to be a universally descending direction.

APPENDIX A PROOFS

a) *Proof of Proposition 1:* The GDA claim is immediate from Eq. (3). On the closed ball $\bar{B}(z; r) \subset \mathcal{D}$, let M_i and B_F bound $\nabla^2 F$ ($i = 1, 2$) and F . For $s \leq \min\{r/B_F, (2M_1)^{-1}, 1\}$, the contraction $T_s(u) = z - sF(u)$

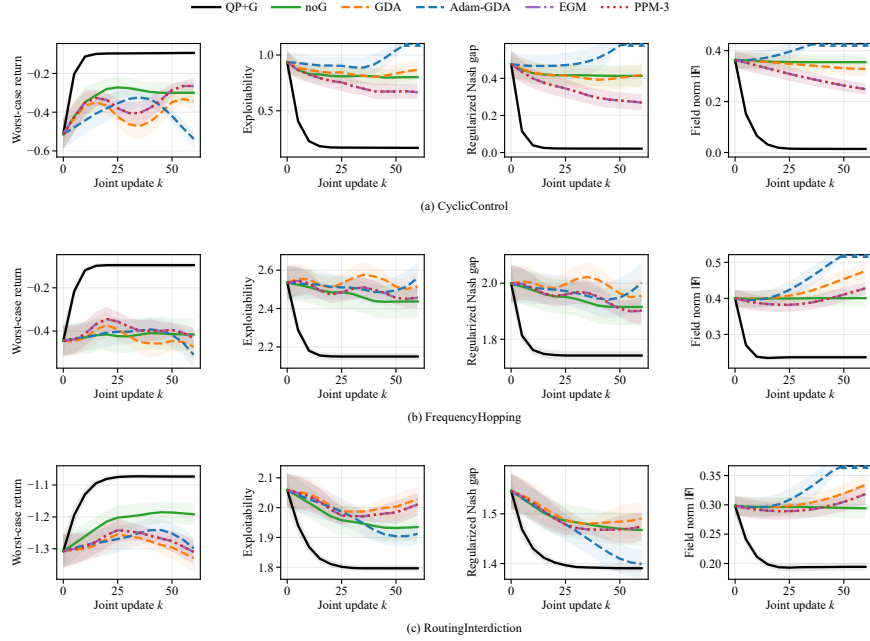


Fig. 4. Neural-policy population-oracle results over ten test seeds (mean $\pm$ standard error); triangles mark clipped Adam-GDA means.

has PPM fixed-point displacement $\|\Delta\| \leq sB_F$. Taylor expansion yields $\Delta + s\mathbf{F} = -s\nabla\mathbf{F}\Delta - s\mathbf{r}_2$, $\|\mathbf{r}_2\| \leq M_2\|\Delta\|^2/2$, and hence $\Delta = -s\mathbf{F} + s^2\nabla\mathbf{F}\mathbf{F} + O(s^3)$ uniformly. For EGM, $\mathbf{F}(z - s\mathbf{F}) = \mathbf{F} - s\nabla\mathbf{F}\mathbf{F} + O(s^2)$ uniformly; multiplication by $-s$ gives the same expansion. ■

b) *Proof of Lemma 1:* With $\mathbf{A} = \nabla\mathbf{F}$ and $\nabla\phi = \mathbf{A}^\top\mathbf{F}$, $\langle\nabla\phi, -\mathbf{F}\rangle = -\mathbf{F}^\top\mathbf{S}\mathbf{F}$ and $\langle\nabla\phi, \mathbf{G}\rangle = \mathbf{F}^\top\mathbf{A}^2\mathbf{F}$. Expanding $\mathbf{A}^2$ gives $\mathbf{S}^2 + \mathbf{S}\mathbf{W} + \mathbf{W}\mathbf{S} + \mathbf{W}^2$; the cross term is antisymmetric, whereas $\mathbf{F}^\top\mathbf{S}^2\mathbf{F} = \|\mathbf{S}\mathbf{F}\|^2$ and $\mathbf{F}^\top\mathbf{W}^2\mathbf{F} = -\|\mathbf{W}\mathbf{F}\|^2$, proving Eq. (6) and Eq. (7). ■

c) *Proof of Theorem 1:* Eq. (7) proves Eq. (8). For (i), $\mathbf{S}\mathbf{F} = 0$ gives $\langle\nabla\phi, \mathbf{G}\rangle = -\|\mathbf{W}\mathbf{F}\|^2 < 0$ when $\mathbf{W}\mathbf{F} \neq 0$, while Eq. (6) gives $\langle\nabla\phi, -\mathbf{F}\rangle = 0$. For (ii), $\mathbf{W} = 0$ gives $\langle\nabla\phi, \mathbf{G}\rangle = \|\mathbf{S}\mathbf{F}\|^2 \geq 0$. For (iii), normality on the invariant plane gives $\mathbf{S} = \mu\mathbf{I}$ and $\mathbf{W} = \sigma\mathbf{J}_0$; hence Eq. (8) is equivalent to $|\sigma| > |\mu|$. ■

d) *Proof of Lemma 2:* Nonnegativity gives $\mathcal{V} \geq 0$; $\mathcal{V} = 0$ forces $\mathcal{E}_F = 0$ and thus $\mathbf{F} = 0$ because $\lambda_F > 0$ and $\mathbf{W}_V \succ 0$. Composition and the chain rule give smoothness and the gradient formula, while $\mathcal{V} \geq \lambda_F\mathcal{E}_F/\kappa_F = \lambda_F\Phi/(2\kappa_F)$ gives (iv). ■

e) *Proof of Lemma 3:* Taylor's theorem on the admissible segment gives Eq. (12) with remainder $\nabla^3\mathcal{V}(\mathbf{z}_k + \theta\Delta_k)[\Delta_k]^3/6$, $\theta \in (0, 1)$; Eq. (13) and $\|\Delta_k\| \leq \beta\|\hat{\mathbf{F}}_k\| + \gamma\|\hat{\mathbf{G}}_k\|$ give the bound. ■

f) *Proof of Proposition 2:* Positive definiteness makes $\hat{q}_k$ strictly convex; solving $\widehat{\mathbf{H}}_k(\beta, \gamma)^\top = (\hat{e}_k, \hat{d}_k)^\top$ gives Eq. (17) and its explicit inverse gives Eq. (18). An infeasible interior point leaves a KKT point on an axis, giving the stated clipped scalar candidates. ■

g) *Proof of Proposition 4:* The comparator sets are subsets of $[0, \infty)^2$, so set inclusion and scalar minimization prove dominance and their formulas. Completing the square in Eq. (18) gives $\hat{\Delta}_k^* = \hat{e}_k^2/(2\hat{c}_k) + (\hat{b}_k\hat{e}_k + \hat{c}_k\hat{d}_k)^2/(2\hat{c}_k\hat{d}_k)$.

Subtracting $[\hat{e}_k]_+^2/(2\hat{c}_k)$ proves Eq. (19) and its equality condition. ■

h) *Proof of Proposition 5:* For $\tilde{\mathbf{A}}(z) \triangleq \mathbf{W}_V^{1/2}\mathbf{A}(z)\mathbf{W}_V^{-1/2}$, define $\tilde{\mathbf{S}}(z) \triangleq \text{sym}(\tilde{\mathbf{A}}(z))$, $\tilde{\mathbf{W}}(z) \triangleq \text{skew}(\tilde{\mathbf{A}}(z))$, and

$$h(z, \mathbf{u}) \triangleq \max \left\{ \mathbf{u}^\top \tilde{\mathbf{S}}(z) \mathbf{u}, \|\tilde{\mathbf{W}}(z) \mathbf{u}\|^2 - \|\tilde{\mathbf{S}}(z) \mathbf{u}\|^2 \right\}.$$

At $\mathbf{z}^*$, a unit $\mathbf{u}$ either has $\mathbf{u}^\top \tilde{\mathbf{S}}(\mathbf{z}^*) \mathbf{u} > 0$, or positive semidefiniteness gives $\tilde{\mathbf{S}}(\mathbf{z}^*) \mathbf{u} = 0$ and nonsingularity gives $\|\tilde{\mathbf{W}}(\mathbf{z}^*) \mathbf{u}\| = \|\tilde{\mathbf{A}}_* \mathbf{u}\| > 0$. Hence $h > 0$ on the compact unit sphere, so continuity yields $c_* > 0$ and a neighborhood $\mathcal{N}_*$ where $h \geq c_*$ uniformly. For $\Phi(z) > 0$, set $\mathbf{u} = \mathbf{W}_V^{1/2}\mathbf{F}(z)/\sqrt{\Phi(z)}$; then $\|\mathbf{u}\| = 1$, $\chi_F/\Phi = \mathbf{u}^\top \tilde{\mathbf{S}} \mathbf{u}$, and $\chi_G/\Phi = \|\tilde{\mathbf{W}} \mathbf{u}\|^2 - \|\tilde{\mathbf{S}} \mathbf{u}\|^2$, which proves the result. ■

i) *Proof of Lemma 4:* Select the larger signal in Eq. (11), lower-bound its performance term by $-\lambda_P B_{P,D}/\kappa_P$, and apply Eq. (20) to obtain Eq. (21). ■

j) *Proof of Proposition 6:* Subtract the models and use $\hat{c}_k = \tilde{c}_k + \rho_{c,k}$ and $\hat{a}_k = \tilde{a}_k + \rho_{a,k}$. In $q_k - \hat{q}_k$ the shifts are nonpositive; dropping them and applying the five coefficient-error bounds gives the first inequality in Eq. (23), while reversing the difference gives the comparator bound. ■

k) *Proof of Lemma 5:* Eq. (14), Taylor's bound, and the definitions of $\varepsilon_k^{\text{model}}$ and $\varepsilon_k^{\text{step,box}}$ successively give Eq. (24). ■

l) *Proof of Lemma 6:* The step caps and $(a+b)^3 \leq 4(a^3 + b^3)$ imply $\varepsilon_k^{\text{rem}} \leq (2M_{3,k}/3)(\beta_{\max} + \gamma_{\max})^3(\|\hat{\mathbf{F}}_k\|^3 + \|\hat{\mathbf{G}}_k\|^3)$. Conditional expectation gives Eq. (25), and the uniform bounds give its stated specialization. ■

m) *Proof of Lemma 7:* Fix $k \geq k_0$ and use comparator $(s, 0)$ for $j_k = F$ or $(0, s)$ for $j_k = G$. Box optimality, Lemma 5, and Proposition 6 bound the realized increment by the comparator model plus $\varepsilon_k^{\text{box}}$ and, respec-

tively, $s\delta_{e,k} + s^2(\delta_{c,k} + \rho_{c,k})/2$ or $s\delta_{d,k} + s^2(\delta_{a,k} + \rho_{a,k})/2$. Conditional expectation of the linear term contributes at most $-sD_{j_k}(z_k) + sG_V\bar{b}_D$, while $\mathbb{E}[\|\hat{\mathbf{F}}_k\|^2 \mid \mathcal{F}_k] \leq M_F$ and $\mathbb{E}[\|\hat{\mathbf{G}}_k\|^2 \mid \mathcal{F}_k] \leq M_G$ bound the Hessian terms. Collecting the shift, coefficient-error, and residual budgets gives $\mathcal{R}_D(s)$ in Eq. (27). ■

n) *Proof of Theorem 2:* At each k , choose the $\mathcal{F}_k$ -measurable selector attaining $\max\{D_F(z_k), D_G(z_k)\}$. Lemma 4 and Eq. (27) give $\mathcal{D}_k \leq -s\mu_{\text{geo}}\Phi(z_k) + \mathcal{R}_{\text{geo}}(s)$. Taking total expectation, summing from k_0 to $K-1$, using $\mathcal{V} \geq 0$, and dividing by $s\mu_{\text{geo}}(K - k_0)$ proves Eq. (28). ■

APPENDIX B

FINITE MARKOV-GAME ENVIRONMENTS

Let S denote the number of states; all games use actions $a, b \in \{0, 1, 2\}$ and a uniform initial state. For $S > 1$, $P(s' \mid s, a, b) = (1 - \xi)\mathbf{1}_{\{s'=T(s,a,b)\}} + \xi\mathbf{1}_{\{s' \neq T(s,a,b)\}}/(S-1)$, where T, ξ , and $\mathbf{1}$ denote the nominal next-state map, transition-noise probability, and event indicator. Let $\varphi_s = 2\pi s/S$ and $M \triangleq \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$. RPS has $S = 1$, reward M_{ab} , and a deterministic self-loop; the remaining games are: *CyclicControl*: $S = 5$, $\xi = 0.02$, $T = (s + 1 + ((a - b) \bmod 3)) \bmod S$, and $r_s(a, b) = (1 + 0.12 \cos \varphi_s)M_{\pi_s(a), \pi_s(b)}$, where $\pi_s(j) = (j - s) \bmod 3$. *FrequencyHopping*: $S = 8$, $\xi = 0.08$, $T = (s + 1 + a - b) \bmod S$, $q_s(a) = 0.25 \cos(\varphi_s + 2\pi a/3)$, and $r_s(a, b) = q_s(a) + 0.55$ if $a \neq b$, while $r_s(a, a) = q_s(a) - 1.10$. *RoutingInterdiction*: $S = 6$, $\xi = 0.10$, $T = (s + 1 + \mathbf{1}_{\{a=b\}} + a) \bmod S$, $c_s(a) = 1 + 0.20 \sin(\varphi_s + 2\pi a/3)$, and $r_s(a, b) = c_s(a) - 0.55 - 1.35\mathbf{1}_{\{a=b\}} - 0.15\mathbf{1}_{\{a \neq b\}}$. *Security-Patrol*: $S = 8$, $\xi = 0.08$, $T = (s + 1 + b + \mathbf{1}_{\{a \neq b\}}) \bmod S$, $v_s(b) = 0.8 + 0.25 \cos(\varphi_s + 2\pi b/3)$, and $r_s(a, b) = v_s(b)$ if $a = b$, while $r_s(a, b) = -0.55v_s(b)$ otherwise. Neural policies use $x_s = (\sin \varphi_s, \cos \varphi_s, u_s, 1)$, with $u_s = s/(S-1)$, range $_a q_s(a)$, std $_a c_s(a)$, or mean $_b v_s(b)$ in the listed order.

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